Lesson plan · for the teacher

Design an AI-supported unit

AI · Level 4 · Lesson 4

Time
40 min

Materials
Board and chalk; A table of the last ten school days: weather, wind, played outside (prepare it, or fill it with the class); Slips of paper, one per day (optional)

Learning goals

- Collect data about real school days in a table.
- Build a decision tree of yes/no questions from our data.
- Test our tree and say when it would be wrong.

Activity: Decision tree from our data

1. Warm-up (5 min): Ask: what decides whether we play outside at break? Collect ideas such as rain, wind, heat and the teacher on duty.
2. Data (8 min): Draw a table with one row per day for the last ten school days and three columns: weather (sunny, cloudy, rainy), windy (yes or no) and played outside (yes or no). Fill it with the class.
3. Build (14 min): Ask which yes/no question best splits the days into “played” and “stayed in”, such as “Is it raining?”. Count the days on each side. Where a branch is still mixed, ask another question. Stop when every branch gives one answer, or when no question is left that splits it.
4. Test (8 min): Use the tree to predict today and two made-up days. Pupils follow the branches aloud and check the prediction against what happened or would happen.
5. Wrap-up (5 min): Ask when the tree would be wrong — for example on a hot day or an exam day — and what extra data would help. Point out that we can read every step of this model. Then give the exit ticket.

Prepare a table of about ten days in case the class can’t remember them; real days are better, but invented ones work. Keep two or three features with simple values. At each branch, count the days that played and stayed in: the best question makes each side as unmixed as possible. If two days have the same weather and wind but different answers, no question can split them: call it a day the tree gets wrong, and ask what else was different (an exam, the teacher on duty).

Discussion

- Which question did our tree ask first, and why?
- When would our tree give the wrong answer?
- Is a model you can read easier to trust than one you can’t? Why?

Exit ticket

- One question our tree asks, and one day when it would be wrong.

Try it in class tomorrow
Build a “Decision tree from our data”: With the class, fill a table of the last ten school days: the weather (sunny, cloudy or rainy), whether it was windy, and whether you played outside at break.

Build a tree of yes/no questions on the board that predicts “play outside”. Test it on today, then ask: when would our tree be wrong?

Worksheet · for pupils

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Name _____ Class _____ Date _____

1 **Half of Mr Okafor's class has no phone at home. How should his unit use AI?**

- ◯ AI homework only for pupils with phones
- ◯ AI work in class, in groups, with a paper path
- ◯ Ask pupils to borrow a phone at home

2 **Put the steps of planning an AI-supported unit in order.**

- ☐ Plan how to judge the unit and share it
- ☐ Decide where AI helps, where pupils use it and where it stays out
- ☐ Plan the activities
- ☐ Decide the evidence of learning
- ☐ Write the learning goals

3 **Match each part of the flood unit to how it is planned.**

Rainfall worksheets at three levels	• • • •	A short quiz before and after
Flood-safety tips	• • • •	No AI: the reading is the learning
Reading the class rain gauge	• • • •	Groups ask AI, then check the official leaflet
Judging whether the unit worked	• • • •	AI drafts; the teacher checks

4 **Complete the sentence.**

Plan from the _____, give AI a place only where it _____, and give every AI activity a path for pupils with no _____.

Word bank: goals · helps · device · tools · marks

✎ **Choose a unit you will teach next term. Where will AI help you, where will pupils use it, and where will it stay out?**